Title: Qubit Decoherence Suppression in Quantum Error Correction: Lessons from Variational Quantum Eigensolvers

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Authors: N. Haddad, W. Dubois, T. Chen, K. Kowalski (Institute of Data and Intelligent Systems, Summit College)

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Abstract:

Qubit decoherence suppression remains a critical challenge for quantum error correction, particularly in quantum chemistry simulation. This paper proposes a framework built on variational quantum eigensolvers that explicitly accounts for scalable qubit connectivity. We evaluate the framework on simulated Hamiltonians of H2 and LiH molecules and compare it against 3 baseline methods. Experimental results show that the proposed approach improves approximation ratio by 33.8% while keeping logical error rate within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of quantum error correction has changed how researchers and practitioners approach qubit decoherence suppression, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, variational quantum eigensolvers have attracted sustained attention as a promising route toward solving qubit decoherence suppression in realistic settings such as quantum chemistry simulation. A central difficulty is that improvements in approximation ratio frequently come at the expense of logical error rate, forcing designers to accept trade-offs that are poorly understood. This paper addresses these gaps by studying how variational quantum eigensolvers interact with superconducting transmon qubits when applied to qubit decoherence suppression.

Our work is motivated by the growing demand for quantum chemistry simulation, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of qubit decoherence suppression, a practical design built on variational quantum eigensolvers, and an empirical evaluation on simulated Hamiltonians of H2 and LiH molecules. Understanding scalable qubit connectivity is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Mukherjee et al. (2025) introduced an early framework based on variational quantum eigensolvers, reporting a 44.8% improvement in approximation ratio over previous baselines. Subsequent work by Larsen et al. (2016) extended this line of research to quantum chemistry simulation, although the evaluation was restricted to small-scale settings. A survey by Zhang et al. (2018) categorised existing approaches to qubit decoherence suppression into three families and highlighted the lack of standard benchmarks. Researchers working on quantum error correction have explored superconducting transmon qubits as an alternative, with Larsen et al. (2018) noting that the approach scales poorly beyond 16 components. More recently, Müller et al. (2009) combined variational quantum eigensolvers with superconducting transmon qubits, achieving a 14.4% reduction in overhead on simulated Hamiltonians of H2 and LiH molecules.

Several studies have criticised the reliance on synthetic workloads, arguing that results on simulated Hamiltonians of H2 and LiH molecules may not generalise to quantum chemistry simulation. In contrast to these efforts, our work treats qubit decoherence suppression and scalable qubit connectivity jointly rather than

as independent concerns. Industry reports corroborate these academic findings, pointing to qubit decoherence suppression as a persistent obstacle for quantum chemistry simulation. Despite this body of literature, no prior study has systematically compared variational quantum eigensolvers and superconducting transmon qubits under identical experimental conditions.

3. Methodology

We formalise qubit decoherence suppression as a constrained optimisation task in which the objective is to maximise approximation ratio while keeping gate fidelity within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on variational quantum eigensolvers, and a validation layer that incorporates superconducting transmon qubits. All experiments were conducted on simulated Hamiltonians of H2 and LiH molecules, which provides 5000 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 10 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies variational quantum eigensolvers to reduce the effect of scalable qubit connectivity, followed by a calibration step that stabilises circuit success probability. We compare against 3 baseline methods, including a conventional approach built on superconducting transmon qubits and two recent variants reported in the literature. The proposed design improves approximation ratio by 33.8% relative to the strongest baseline and reduces approximation ratio by 35.0%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the superconducting transmon qubits component reduces approximation ratio by approximately 7.5%, confirming its contribution. A sensitivity analysis over 8 parameter settings shows that performance remains stable, with variation in circuit success probability below 2.2%. Threats to validity include the limited diversity of simulated Hamiltonians of H2 and LiH molecules and the possibility that deployment in quantum chemistry simulation introduces conditions not captured in our setup.

4. Conclusion

This paper investigated qubit decoherence suppression through the lens of variational quantum eigensolvers, and demonstrated that a carefully designed pipeline can improve approximation ratio by 33.8% on simulated Hamiltonians of H2 and LiH molecules. The findings suggest that quantum error correction will continue to benefit from closer integration of variational quantum eigensolvers and superconducting transmon qubits, particularly in quantum chemistry simulation. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of scalable qubit connectivity. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for qubit decoherence suppression.